

## EXPERIMENT NO: 6

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**Subject Name:** System Design

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### Aim

To design a highly scalable video streaming platform similar to YouTube that supports adaptive bitrate streaming (HLS/DASH), dynamic manifest handling and real-time engagement features (likes and comments) with high availability and low latency.

### Objectives

- Understand Adaptive Bitrate Streaming using HLS and DASH protocols.
- Implement scalable like and comment systems videos.
- Ensure fault tolerance, high availability, and eventual consistency.

### Functional Requirements

1. Users can sign up and log in using their emails.
2. Users can upload videos.
3. Users can watch videos.
4. Users can like/dislike videos.
5. Users can comment on videos.
6. Users can search videos.
7. System should support multiple resolutions (360p, 720p, 1080p).

### Non-Functional Requirements

1. Scalability: 100M+ DAU
2. CAP Theorem: Availability >>>> Consistency

Users should always be able to watch videos

Target: 99.99% uptime

3. Latency: 50-80 ms

User should be able to see the videos with zero or negligible buffering

## Core Entities of the System

1. User
2. Video
3. Likes and Comments
4. Channel

## API Design

### User-Onboarding

1. POST Call /api/v1/auth/signup
2. POST /api/v1/auth/login

### Upload a Video

POST Call /videos/upload

#### Request

```
{  
  title,  
  description,  
  video_file  
}
```

### Get Video

GET Call /videos/{video\_id}

#### Returns

```
{  
  title  
  description  
  views  
  video_url  
}
```

### Like & Dislike video

1. POST Call /api/v1/videos/{video\_id}/like
2. POST Call /api/v1/videos/{video\_id}/dislike

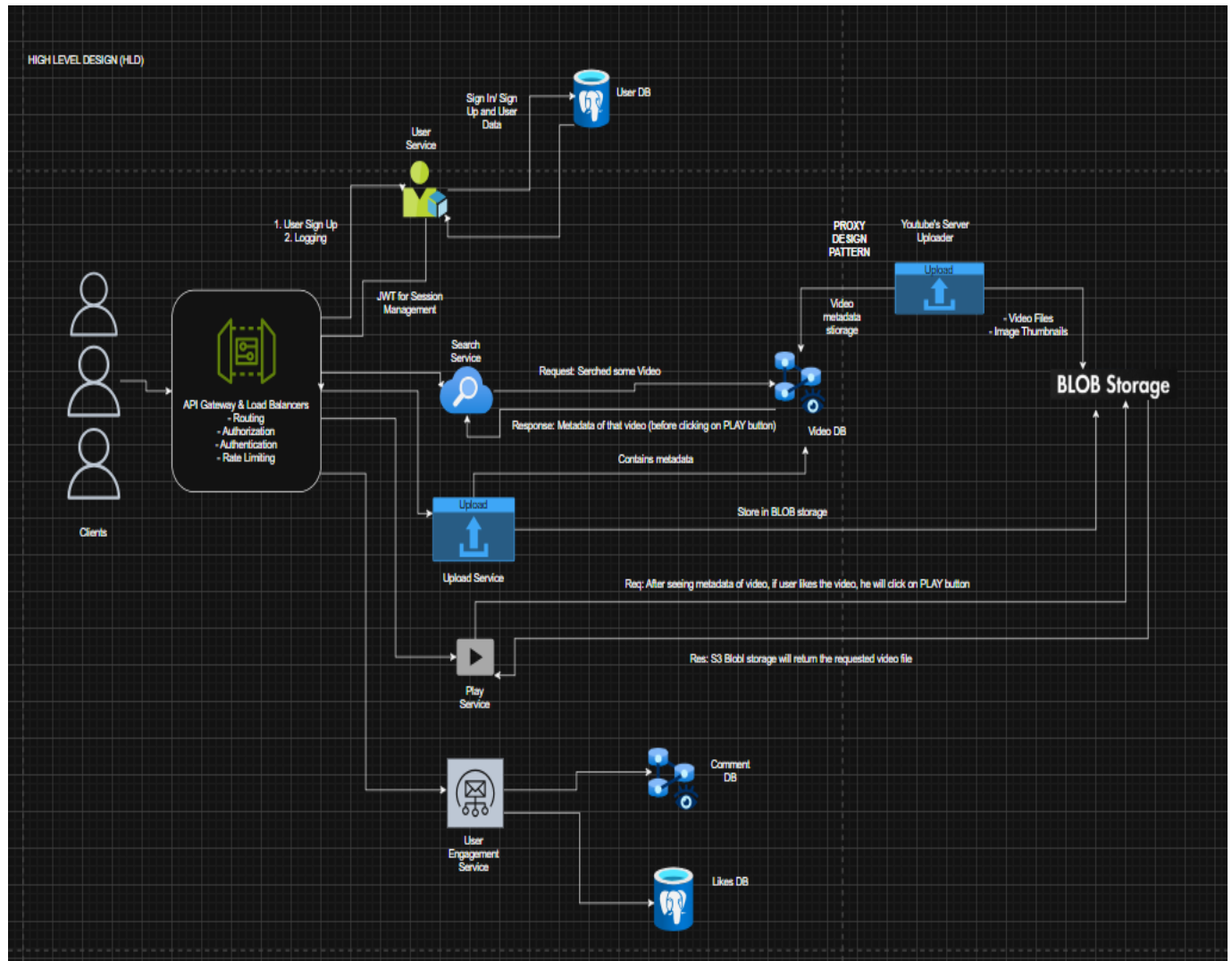
### Comment on Video

POST Call /videos/{video\_id}/comment

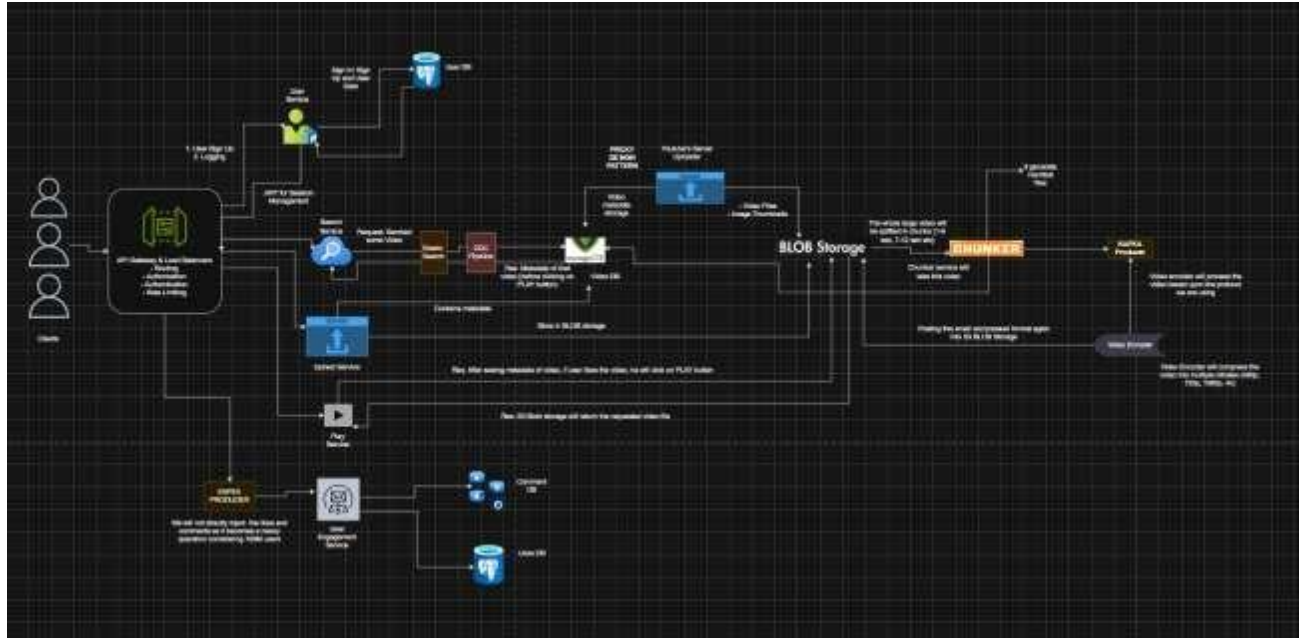
### Search Video

GET Call /videos/search?q=music

## High Level Design (HLD)



## LOW-LEVEL DESIGN (LLD)



## Outcome

- Understand the architecture of a scalable video streaming platform similar to YouTube.
- Learn how adaptive bitrate streaming using HLS and DASH improves video playback under varying network conditions.
- Design scalable APIs and data models to support real-time engagement features such as likes and comments.
- Apply distributed system principles (scalability, high availability, low latency, and eventual consistency) in large-scale applications.